Introduction to Machine Learning

A short sample document for testing PDF text extraction

1. What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence that enables computers to learn patterns from data and use those patterns to make predictions or decisions. Instead of writing explicit rules for every situation, developers train a model using examples.

For example, an email spam filter can learn from messages labeled as spam or not spam. After training, the model can estimate whether a new message is likely to be spam.

2. Main Types of Machine Learning

Supervised learning: The model learns from labeled examples. Common tasks include classification, such as identifying spam, and regression, such as predicting house prices.

Unsupervised learning: The model looks for structure in data without target labels. Clustering customers into groups is one example.

Reinforcement learning: An agent learns by taking actions and receiving rewards or penalties from an environment. It is used in areas such as robotics and game-playing.

3. The Machine Learning Workflow

A typical ML project begins by defining the problem and collecting relevant data. The data is then explored and cleaned. Missing values, duplicates, inconsistent formats, and unusual observations may need attention.

Next, the dataset is divided into training and test sets. The training set is used to fit the model, while the test set helps estimate how well it performs on data it has not seen.

After training, developers evaluate the model with suitable metrics. Accuracy can be useful for some classification tasks, but precision, recall, and F1-score may be more informative when classes are imbalanced.

4. Important Terms

Dataset: A collection of examples used for analysis or model development.

Feature: An input variable used by a model, such as age, attendance, or study hours.

Label: The target value a supervised model learns to predict.

Training: The process of adjusting a model using data.

Overfitting: When a model learns details of the training data too closely and performs poorly on new examples.

5. Example: Predicting Student Performance

Imagine a college wants to understand which factors are associated with student exam marks. A dataset might include attendance percentage, weekly study hours, previous marks, and final marks.

A regression model could use attendance, study hours, and previous marks to estimate final marks. The estimate is not guaranteed to be correct. Model quality depends on the data, the chosen method, and evaluation on examples that were not used during training.

6. Questions for Practice

1. What is the main purpose of machine learning?

2. How does supervised learning differ from unsupervised learning?

3. Why should a dataset be divided into training and test sets?

4. What does overfitting mean?

5. Name two metrics that can be used to evaluate a classification model.

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